



INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Introduction to Cryptography

Lab Session 1: Modular arithmetic

September 5, 2025

Please solve the following exercises by your own although collaboration with other student is allowed. Develop the programming exercises in C/C++ programming language. **At the end of the session you must present advances for exercises 1 and 2.**

1. Programming exercises

- Design and implement a function that receives an integer $n \geq 2$, and print or send to a file the multiplication table for $\mathbb{Z}_n - \{0\}$, i.e., only print the product for elements $a, b \in \{1, 2, \dots, n-1\}$, remember that to do the product, you must do $a*b \bmod n$. Test your function generating at random the following values for n :
 - Even numbers for $2 \leq n \leq 50$.
 - Odd numbers for $2 \leq n \leq 50$.
 - Prime number for $2 \leq n \leq 50$. Look for a list of prime numbers in the internet and use it.

Analyze the table that you generate, see for different values of n if every element in $a \in \mathbb{Z}_n$ has multiplicative inverse, i.e. $b \in \mathbb{Z}_n$ such that $a * b \bmod n = 1$. What can you conclude?

- Implement the shift cipher considering the following requirements:
 - You must code and decode every symbol in the plaintext/ciphertext with an element of $\mathbb{Z}_n$, where n is the size of the alphabet.
 - Use addition and/or subtraction in $\mathbb{Z}_n$ to perform enciphering and deciphering.
 - Implement a function to encipher, this function must receive the plaintext M and K as input, and print the ciphertext C .
 - Implement a function to decipher, this function must receive the ciphertext C and K as input, and print the plaintext M .
- Test your implementation of the shift cipher using messages with at least 50 characters to encipher and decipher. Test your program with the following alphabets
 - Use the English alphabet, i.e. the size of the alphabet will be $n = 26$.
 - Use the (character code 32-127) as alphabet. Determine correctly the size of this alphabet.
 - Ask for the user the size of the alphabet n , and choose at random n characters from the ASCII printable characters to have the alphabet.

2. Products

Every student must write a brief report, containing at least:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Your conclusion for exercise 1. Justify your answer showing relevant results of your test.
3. The source code for points 2 and 3 and a brief explanation of how you implement the function.
4. Screen shots of your programs running.

3. Evaluation

- Advances in class (Exercises 1 and 2): 3 points
- Source code: 2 points
- Program running: 3 points
- Written report: 2 points